Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

Exercise 1: Choice of realized variance estimator

We first load the high-frequency minute price data and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp.

The data contain minute-level closing prices during regular trading hours. We use a fixed one-minute calendar-time grid from 09:30 to 15:59 for each trading day. This corresponds to 390 one-minute observations per normal trading day. Since a stock may not trade in every minute, we handle missing prices using previous-tick interpolation within each trading day only. This means that if AAPL has no observed price in a given minute, we carry forward the most recently observed price from the same trading day. We do not carry prices across trading days, and we do not backfill prices before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, we sample prices on a fixed grid relative to the market open. For example, with 10-minute sampling, prices are taken at 09:30, 09:40, 09:50, and so on. This differs from a bucket-based approach, where the last observed price within each Δ-minute interval is used. We prefer the fixed-grid approach because it makes the sampling rule explicit and because it will later allow us to align intraday returns across all 30 stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where (r_{d,j}^{( )}) is the intraday log return over sampling interval.

As shown in Figure 1, average realized variance is highest at the shortest sampling intervals and decreases as the sampling interval increases. This pattern is consistent with market microstructure noise at very high frequencies, such as bid-ask bounce, price discreteness, and stale prices. At very fine sampling intervals, these frictions can inflate realized variance because observed transaction prices deviate from the latent efficient price. Sampling less frequently reduces this noise, but very long sampling intervals also discard useful intraday information.

References

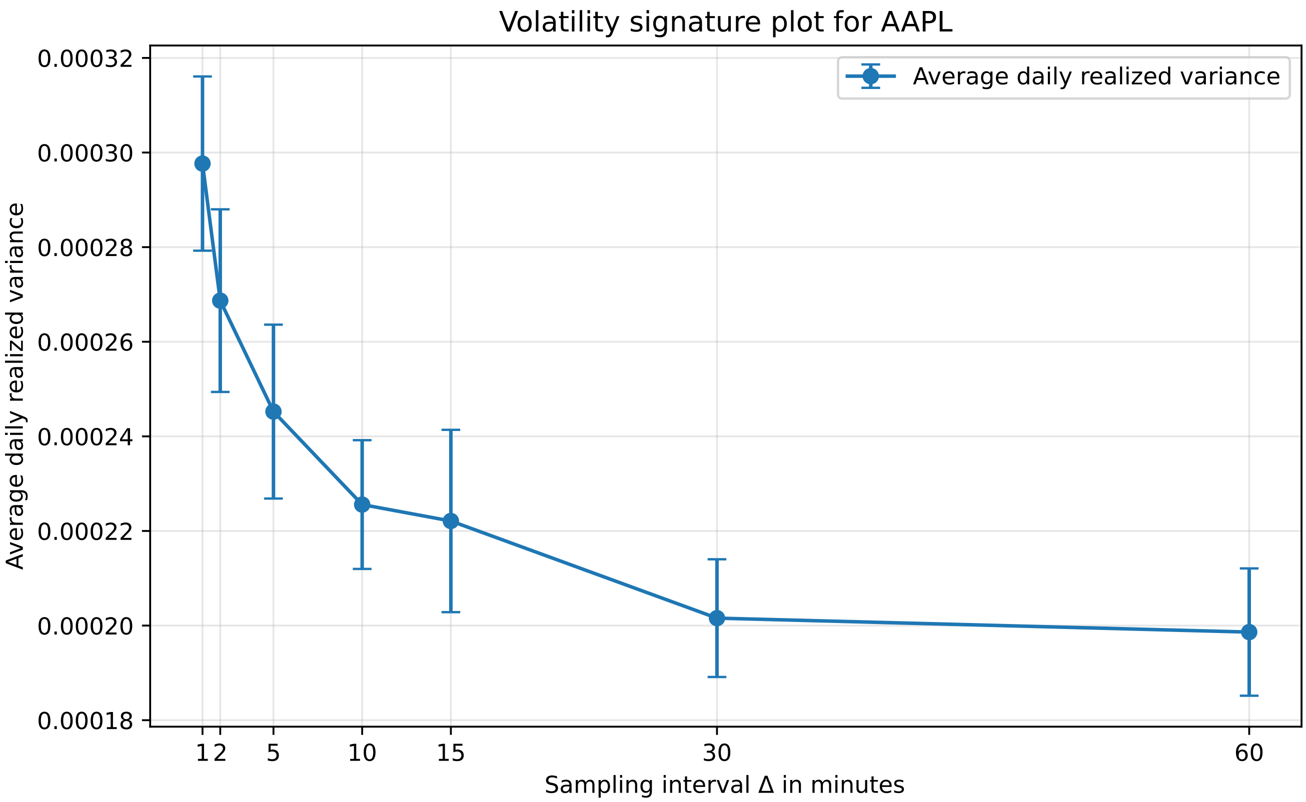

Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.